

Chapter 5

Optical Techniques for Systems with Competing Interactions

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Abstract Optical spectroscopy and Raman scattering can be used to probe all the relevant degrees of freedom in a solid. In this chapter, we present an overview of the elementary and collective excitations accessible in compounds where spin, orbital, and lattice degrees of freedom compete strongly. For illustration, we will focus in particular on the magnetic response of the weakly coupled spin-tetrahedron system $\text{Cu}_2\text{Te}_2\text{O}_5(\text{Br},\text{Cl})_2$, on optical phonons in the highly frustrated pyrochlore antiferromagnets CdCr_2O_4 and ZnCr_2O_4 , and on exciton-magnon excitations in the quasi-one-dimensional spin-1/2 Heisenberg antiferromagnet KCuF_3 .

5.1 Introduction

In an inelastic light-scattering process, monochromatic light is scattered by variations in the electronic polarizability. The involvement of virtual electron-hole pairs leads to a strong sensitivity to local lattice distortions, as exemplified by local orbital states in manganites, or to local magnetic exchange processes in Heisenberg exchange-coupled systems. While magnetic light scattering in long-range-ordered systems leads to two-magnon signals whose shape is reminiscent of a density of states, frustrated spin topologies and topological ordering may lead to sharper, quasiparticle-like modes. Indeed, inelastic light-scattering has a very high sensitivity to singlet excitations, for example in weakly coupled spin tetrahedra. However, depending on the exchange geometry of the system and the temperature of the experiment, a multitude of overlapping modes, or quasielastic scattering, has also been observed [1, 2]. Inelastic scattering by electronic transitions is allowed at low energies if the scattering rate is large and screening can be prohibited. A number of specialized reviews have been written which cover correlated electron systems [3], high-temperature superconductors [4, 5], two-leg spin ladders [6, 7], and the materials aspects of perovskites [8].

In infra-red (IR) optical spectroscopy, the properties of a system are probed through the intensity and the polarization state of the reflected, absorbed, or transmitted light over an extended frequency range. The dominant contribution to

the excitation processes observable by IR spectroscopy are transitions which are electric-dipole-allowed (ED-allowed, e.g. IR-active phonons), and hence the spin degrees of freedom are generally considered not to be directly accessible by IR spectroscopy. Magnetic-dipole (MD) transitions, however, are optically allowed and can be observed at far-infra-red (FIR) frequencies in magnetically ordered systems. This one-magnon excitation process is called antiferromagnetic resonance (AFMR) and can provide valuable information on the symmetry of a magnetically ordered phase. Because MD transitions are several orders of magnitude weaker than ED transitions, their detection is more probable in transmission geometry than in reflection. Recently, the prototypical antiferromagnetic insulator CoO, which has the rock-salt structure, was investigated in the FIR regime in reflection geometry and ED-active phonons, MD transitions, and electric quadrupole transitions were detected [9]. In addition to these single-ion excitations, cooperative excitations can occur which involve simultaneous transitions on two neighboring exchange-coupled magnetic ions: these include two-magnon absorptions, multi-magnon-plus-phonon processes, exciton-magnon, and exciton-exciton transitions [10].

In the following we will first describe general aspects of and typical set-ups for light-scattering experiments and IR spectroscopy. Then we will discuss examples of Raman and IR experiments where the spin degrees of freedom are clearly detectable in the excitation spectra.

5.2 Inelastic Light-Scattering

In an inelastic light-scattering experiment, monochromatic light in the visible range, for example with the wavelength $\lambda = 532.1$ nm, and with well-defined polarization is focused on the sample surface. Inelastically scattered light with a “Raman shift” (frequency-shift) of $5\text{--}3,000\text{ cm}^{-1}$ is emitted, which has dipole characteristics and is sampled by a photo-lens of sufficiently large aperture. The Raman shift is an energy-proportional scale based on the inverse wavelength, on which the incident light given above corresponds to $18,794\text{ cm}^{-1}$ and numbers on the order of 300 cm^{-1} correspond to a typical phonon energy scale in a transition-metal oxide.

A second lens with an adapted focal length is used to focus the beam of parallel light on the entrance slit of the Raman spectrometer. The two lenses between the sample chamber or cryostat and the spectrometer allow for flexibility with respect to the distance between the experimental components. Figure 5.1 shows a typical set-up for Raman scattering.

Neither sensitivity nor frequency resolution are limiting factors in present-day Raman-scattering experiments. CCD arrays (back-illuminated and liquid-nitrogen-cooled) with long sampling times (20 s–20 min.) and multiple accumulations to remove spurious signals from cosmic muons allow the simultaneous acquisition of frequency shifts in a broad window corresponding to excitations of quasi-particles in the solid. The spectral resolution, which is of order several cm^{-1} ($\equiv 1.44\text{ K} \equiv 0.125\text{ meV}$), depends on geometrical factors including the focal length,

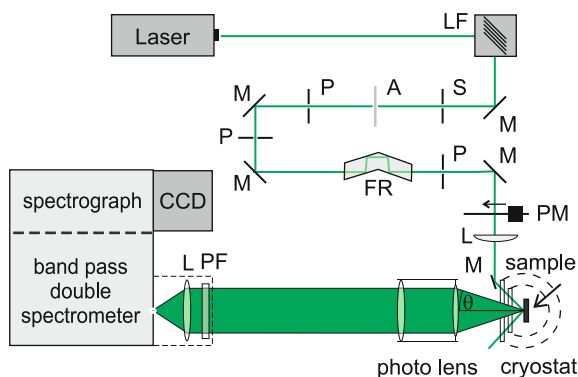


Fig. 5.1 Schematic representation of a typical Raman set-up with mirrors (M), pinholes (P), holographic laser filter (LF), lenses (L), power meter (PM), fresnel rhombus (FR), polarization filter (PF), variable slit (S), and attenuator (A). The triple spectrometer includes a subtractive double monochromator that is used as a band-pass filter and a single monochromator that disperses the frequency band of inelastically scattered light on the CCD detector. The *thick line* denotes the light path from the laser to the sample and from the sample to the spectrometer

the slit widths, the number of grooves on the grating, and the density of pixels on the CCD array.

A limiting factor, however, is the set-up- and sample-related background level of fluorescence and stray light. Its origin may be surface contaminations and defects. As a consequence, freshly cleaved single-crystal surfaces are desirable. Further sources of background contributions include the glass of the collecting lenses, other optical components, and the laser itself. Intrinsic fluorescence processes are observed only rather rarely in extended solid states due to their quenching by nonradiative decay processes.

Taking into account that the scattering cross-section of magnetic and electronic excitations is one to two orders of magnitude smaller than the scattering intensity of phonons, and that these contributions have both low energy and a large bandwidth, the suppression of scattering with a similarly broad energy distribution is essential. For the laser, a pre-monochromator or holographic filter is used. Optical pinholes and a long beam path from the laser to the focussing lens further decrease background radiation. The optical beam path and the optical elements are separated for the exciting and for the scattered light.

Finally a controllable suppression of the elastically scattered light from the sample is gained by using a subtractive double monochromator in front of the dispersing spectrometer. Including the dispersing (“spectrograph”) stage, the spectrometer is then referred to as a triple spectrometer. Single monochromator stages with super-notch filters, as for example in a high-aperture Raman microscope, do not generally comply with these requirements and are more suitable for material analysis and phonon studies.

5.3 Inelastic Phonon Light-Scattering

Optical phonons with $q \approx 0$ provide very large scattering cross-sections. Further, they are easily observable in transition-metal oxides, as the exciting light and the corresponding electron-hole pairs are generally in close proximity to a resonance to interband or charge-transfer (CT) excitations. The factor-group analysis, i.e. the determination of Raman-active phonon modes and their selection rules, can follow different schemes. An introduction may be found in text books on light scattering [11, 12]. The Bilbao Crystallographic server provides web pages with corresponding services [13, 14].¹

Optical phonons and the related selection rules are sensitive probes for lattice instabilities or anomalies whose origin lies in spin–phonon or electron–phonon coupling to fluctuating electronic or magnetic degrees of freedom. In such systems, anomalies of the phonon line width, such as asymmetric phonon lines (Fano line shape) and changes in the slopes of the temperature-dependent frequency or intensities, are very common observations. Quite generally, nature prevents energetic degeneracies and competing interactions of electronic or magnetic origin by using phonon degrees of freedom in a dynamic or a static way. The results are frequency shifts and intensity anomalies as a function of temperature, as represented in Fig. 5.2. Dynamic phonon anomalies spanning large temperature ranges are obviously of most interest [15].

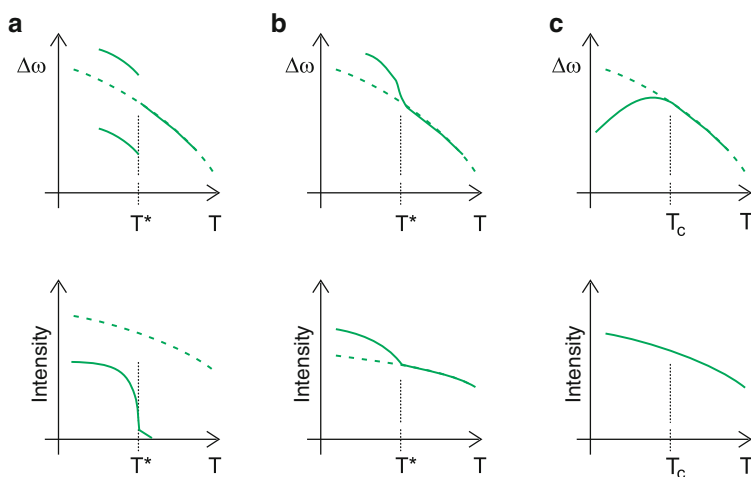


Fig. 5.2 Phonon effects in Raman shift (*upper graph*) and intensity (*lower graph*) which may be observed in different compounds. Dashed lines correspond to the unperturbed situation. Full lines correspond to observations in a system with (a) a structural instability, (b) a crossover due to the mutual coupling of spin and phonon systems, and (c) orbital dynamics or other fluctuating states. In (c), T_c represents the Curie temperature, for example of a weakly doped manganite such as $(\text{La,Sr})\text{MnO}_3$, which precedes the formation of orbital polarons

¹ <http://www.cryst.ehu.es/rep/sam.html>.

Table 5.1 Phonon anomalies in Raman scattering observed in systems featuring mutual interplay of structural, electronic, and magnetic degrees of freedom, with respective examples

Effect	Example
Spin-Peierls transition	CuGeO ₃ [19, 20]
Structural ordering	SrCu ₂ (BO ₃) ₂ [21, 22]
Structural and charge ordering	NaV ₂ O ₅ [23]
Fluctuation-induced phonon softening	TiOCl, TiOBr [24–26]
Crossover behavior	α -TeVO ₄ [27] KCuCl ₃ [28], TiCuCl ₃ [29, 30]
Random interactions	KCu ₅ V ₃ O ₁₃ [31]
Spin-state fluctuations	[Fe(pmd)(H ₂ O){Au(CN) ₂ } ₂] $\cdot$ H ₂ O [32]
Orbital-polaron formation	(La _{1-x} Sr _x)MnO ₃ [16–18]
Canted orbital polarons	(La _{0.5} Sr _{0.5})MnO ₄ [33]
Charge/spin stripes	(La,Sr) ₂ NiO ₄ [34, 35]
Local distortions	CdCr ₂ S ₄ (Gnezdilov et al., unpublished)

Compounds with incommensurate magnetic ordering often show related phonon anomalies close to the transition temperature. In contrast, doped manganites show a continuous softening of phonon frequencies without a characteristic temperature, which is due to the formation of orbital polarons [16–18]. In the sense of a local electronic defect, this type of anomaly shares a number of aspects with the effect of local polar distortions in a system such as CdCr₂S₄ (Gnezdilov et al., unpublished), which shows magnetocapacitive effects.

Different again is multi-phonon scattering, which is enhanced in the compounds under discussion and is usually non-monotonic in intensity but constant in frequency as function of temperature. This higher-energy scattering very often overlaps with magnetic scattering, reflecting the mixing of degrees of freedom under such conditions. In Table 5.1, several examples are listed which demonstrate the use of phonon modes to characterize the electronic or spin degrees of freedom in systems with competing interactions.

5.4 Inelastic Magnetic, Quasielastic, and Electronic Light Scattering

The study of magnetic excitations is helpful because they probe the typical energy scales of the magnetic system. Reviews of both the basic and the advanced aspects of magnetic light scattering in low-dimensional systems may be found in [1, 15], and for three-dimensional classical spin systems in [36].

The magnetic excitation spectra of compounds with a frustrated exchange geometry of quantum spins show a tendency towards spectral downshift compared to the two-magnon density of states of a long-range-ordered system. If the ground state is nonmagnetic, due for example to the formation of a spin gap in a $S = 1/2$ system, asymmetric, low-energy singlet modes may be observed. Further triplet

modes are observed at similar energy scales. Singlet and triplet modes differ with respect to their behavior in a magnetic field. Similar to two-magnon scattering in gapless systems, singlet modes do not shift in frequency in a magnetic field. Longitudinal magnons (triplet excitations) occupy an intermediate position: they may be observed in long-range-ordered states close to a quantum critical point and thus where the ordered moment is strongly reduced. These modes show the temperature dependence expected for the order parameter, but no shift in a magnetic field.

Systems with helical or incommensurate order often show a multitude of sharp excitations which may overlap partly with the background, which in turn is a broader scattering continuum. This complex line shape is difficult to describe and depends on details of the magnetic ordering. The ordered moments of these systems are nevertheless larger when compared to the types of system discussed above, whence spin fluctuations are less pronounced. In Fig. 5.3, we illustrate schematically the approximate line shapes which may be observed in the parameter space given by the degree of competition between the interactions and the connectivity of the spin system. Examples of specific systems are listed in Table 5.2.

Systems of weakly coupled spin tetrahedra form a well established class of materials in which one may study the effect of competing interactions on magnetic excitations by light-scattering [37]. In Fig. 5.4, we show the crystal structure together with the low-energy excitation spectra of several compounds whose stoichiometry is derived from $\text{Cu}_2\text{Te}_2\text{O}_5(\text{Br},\text{Cl})_2$ [38, 39]. The lone-pair ion Te^{4+} , in combination with halogen and oxygen ions, leads to open structures which form the framework for magnetic exchange with restricted connectivity [1].

While the largest exchange constants are those within the tetrahedra, the most important exchange paths in the spin-tetrahedron compounds consist of halogen–oxygen bonds which couple the tetrahedra of Cu^{2+} ions within the crystallographic ab plane. These interactions can be modified by replacing Cl by Br, leading to a weakening of the in-plane with respect to the out-of-plane coupling. The arrangement of the tetrahedra can also be modified as in $\text{Cu}_4\text{Te}_5\text{O}_{12}\text{Cl}_4$ [39], leading to a three-dimensionally coupled system more amenable (than the coupled-chain and planar materials) to a mean-field description.

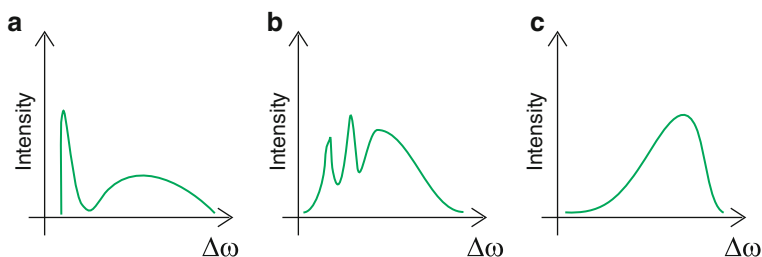


Fig. 5.3 Representation of magnetic light-scattering intensity in different exchange geometries as a function of energy. The spectral-weight distribution depends on the presence of a spin gap at low energy and on the extent of spin frustration. Cases shown are (a) frustrated, dimerized spin chains, (b) coupled spin tetrahedra, and (c) a two-dimensional antiferromagnet

Table 5.2 Examples of magnetic and electronic Raman scattering giving the observed phenomenon, the basic exchange geometry, and example systems. Mode labels are taken primarily from the references given. Mode line widths are determined by an interplay of the system connectivity with the degree of competition between spin exchange constants. This may be characterized by considering the ratio of intra- to inter-chain/dimer/tetrahedron coupling in the spin system. Fluctuations are enhanced as anisotropies increase, making the systems increasingly low-dimensional

Phenomenon	Exchange system	Example
Singlet bound state	Dimerized, frustrated chain	CuGeO ₃ [19]
Longitudinal magnon	Weakly coupled tetrahedra	Cu ₂ Te ₂ O ₅ Br ₂ [37, 38]
Sharp triplets	Coupled tetrahedra	Cu ₂ Te ₂ O ₅ Cl ₂ [39]
Defect-induced bound state	Dimerized, frustrated chain	(Cu,Zn)GeO ₃ [50, 51]
Sharp triplets	Dimerized, coupled chains	(VO ₂)P ₂ O ₇ [52]
Sharp singlets and triplets	Frustrated dimers	SrCu ₂ (BO ₃) ₂ [53–55]
Three-magnon scattering	2D coupled dimers	NaV ₂ O ₅ [56–59]
	Chain, spin-Peierls	CuGeO ₃ [60]
	3D coupled dimers	KCuCl ₃ [28]
Two-magnon continuum	2D AF with stripes	La ₂ NiO ₄ [61]
Two-magnon continuum	3D AF	KNiF ₃ [36, 62, 63]
Continuum, smaller line width	3D coupled dimers	(Ti,K)CuCl ₃ [28–30]
Resonance RS, gap formation	Spin ladder	Sr ₁₄ Cu ₂₄ O ₄₁ [7, 18, 64]
Resonance RS	Square lattice	undoped HTSC [65, 66]
Quasi-elastic Lorentzian	Alternating chain	(VO ₂)P ₂ O ₇ [52]
	Chain, spin-Peierls	CuGeO ₃ [67]
	Chain with helical order	NaCu ₂ O ₂ [68]
Electronic scattering	1D collision-dominated	BaVS ₃ [69]
	2D square lattice	HTSC [3]
	2D triangular lattice	Na _x CoO ₂ [49]
	3D polaronic	EuO, (Eu,La)B ₆ [70]

These systems of weakly coupled spin tetrahedra are found to be at the boundary between a localized product ground state of spin singlets and helical ground states with large unit cells and reduced ordered moments. It has been suggested that this crossover can be parameterized by considering the ratio of inter-tetrahedron coupling and a Dzyaloshinskii–Moriya interaction which is antisymmetric with respect to intra-tetrahedron exchange [38].

All known spin-tetrahedron compounds realize long-range ordering with critical temperatures $T_c < 20$ K [39]. However, the magnitudes of the ordered moments, the ordering vectors, and the energy scales in the inelastic excitation spectra differ strongly. As shown in Fig. 5.4a–c, the low-energy spectrum consists essentially of magnetic excitations with a well-defined and distinctive temperature-dependence. The frequency window of magnetic excitations is approximately 30–80 cm^{−1} for all systems. In addition, a low-frequency “collective mode” and quasielastic scattering are observed for frequencies below 25 cm^{−1}. In the compound with the smallest T_c , and presumably the smallest in-plane inter-tetrahedron interaction, a low-energy mode has been identified as a longitudinal magnon. This excitation follows the

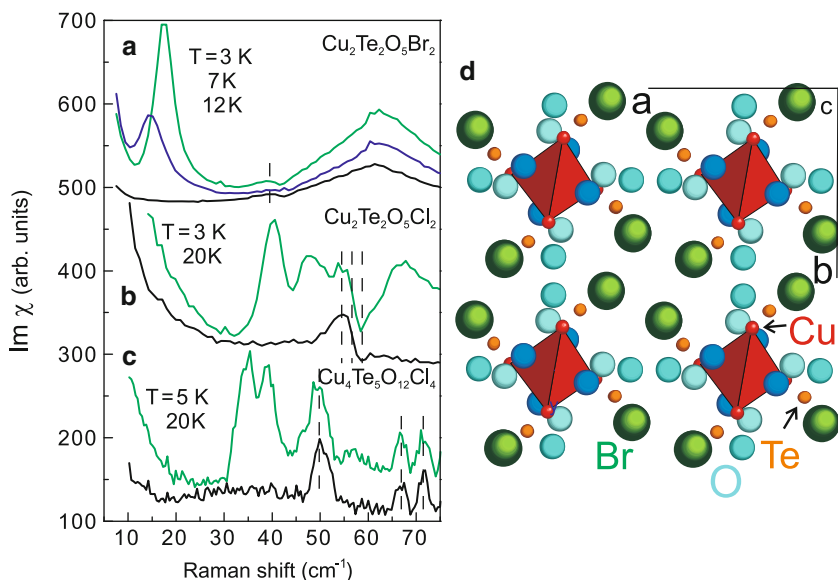


Fig. 5.4 Low-energy Raman scattering in three different compounds with coupled spin tetrahedra at temperatures above and below an ordering temperature T_c . Data are shown for (a) $\text{Cu}_2\text{Te}_2\text{O}_5\text{Br}_2$ ($T_c = 11.4\text{ K}$), (b) $\text{Cu}_2\text{Te}_2\text{O}_5\text{Cl}_2$ (18.2 K), and (c) $\text{Cu}_4\text{Te}_5\text{O}_{12}\text{Cl}_4$ (13.6 K); the curves are shifted for clarity. Dashed lines denote scattering intensity due to phonons. (d) Representation of the crystal structure in ab -plane projection. The tetrahedra formed from Cu ions coordinated by oxygen ions on two different sites are filled. Exchange paths between the tetrahedra are formed by Br (large circles) and O ions (intermediate-sized circles)

temperature-dependence of the order parameter, i.e. of the sublattice magnetization. By contrast, the higher-energy continuum shows no shift and only a depression of its intensity with increasing temperature (Fig. 5.4a). Increasing the in-plane coupling suppresses this mode completely. Instead, a larger number of excitations with smaller line width appears. The more three-dimensional system $\text{Cu}_4\text{Te}_5\text{O}_{12}\text{Cl}_4$ shows only two very sharp modes which have no temperature-dependence of their energy (Fig. 5.4c). This effect is a consequence of the suppression of spin fluctuations in the latter system.

For two- or three-dimensional compounds with larger connectivity and ground-state degeneracy, a multitude of singlet states leading to a continuum of scattering is expected. With the exception of Raman scattering, these singlet excitations can only be observed through their contribution to the specific heat [40]. Examples where such excitations have been detected definitively in Raman studies remain rather rare, although good candidates are compounds with square planar lattices featuring additional diagonal coupling (J_1 – J_2 systems [41]) and kagomé materials [42, 43]. The theoretical relevance of Raman scattering in kagomé systems is worth noting because it can distinguish spin-liquid from symmetry-broken ground states by observing the anisotropy of the magnetic scattering [44] and its line shape at low energies (Läuchli and Lhuillier, unpublished).

We include in this section a discussion of quasielastic and electronic scattering. Because all Raman-scattering processes are by definition electronic, we follow the pragmatic approach of defining as quasielastic scattering any very low-energy scattering which arises due to energy-density fluctuations of the magnetic system. These processes are mediated by spin–phonon coupling and lead to a Lorentzian line width and a temperature-dependence related to the magnetic specific heat [45, 46]. This type of scattering is observed with different intensities in all magnetic compounds where competing interactions are present.

Electronic Raman scattering with a single scattering time has the same Lorentzian line shape and is screened at finite frequencies [5]. However, if the fluctuations consist of variations of the effective mass within an anisotropic band or if sources of anomalous scattering exist, then finite-energy scattering is observable. While the first case was demonstrated first for doped semiconductors under the application of anisotropic pressure [47], the second is relevant in systems such as dirty or correlated metals [48]. In non-Fermi-liquid systems, an anomalous energy- and frequency-dependent scattering rate, arising from polaronic effects or from nesting, leads to a frequency-independent scattering continuum that extends to rather high energies. The best-known candidates for such scattering are high temperature superconductors (HTSC), where the electronic Raman continuum extends to several thousand wavenumbers [3], and the hydrated cobaltates, $\text{Na}_x\text{CoO}_2 \cdot y\text{H}_2\text{O}$ [49], where this contribution is observed in Raman shifts up to 550 cm^{-1} (Table 5.2).

5.5 The IR Experiment

A typical experimental set-up for performing IR measurements is shown in Fig. 5.5. Before impinging on the sample, the light is passed through a Michelson interferometer unit where the beam is split by a beamsplitter (BS) and reflected by a fixed and a scanning mirror (FM and SM). The result is an interferogram which provides the desired spectral information after Fourier transformation.

The shape of the resulting spectrum depends on the emission of the source, and is influenced by all of the optical elements in the beam path. The reflectivity of a sample must thus be obtained as the ratio of its spectrum in comparison to the spectrum of a reference material such as gold, silver, or aluminum, which are assumed to be ideally reflective in a certain frequency range. Extrapolating the measured reflectivity or transmission data to zero and to infinite frequency allows one to perform a Kramers–Kronig transformation to obtain the real and imaginary parts of the dielectric function or the optical conductivity. A direct experimental determination of both imaginary and real parts of the dielectric function becomes possible by performing ellipsometry measurements, i.e. by measuring the exact polarization state of the elliptically polarized light reflected from the sample [71].

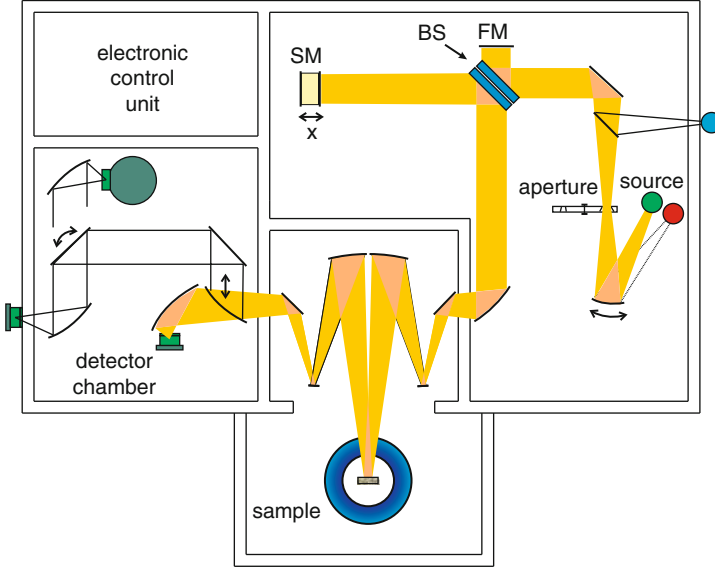


Fig. 5.5 Representation of an IR Fourier spectrometer (Bruker IFS66v) with a Michelson interferometer unit consisting of a beam-splitter (BS), a scanning (SM), and a fixed mirror (FM) [73]

5.6 Spins, Phonons, and Light

To illustrate how electromagnetic radiation with electric- and magnetic-field vectors $\mathbf{E}$ and $\mathbf{H}$ couples to the lattice and spin degrees of freedom, we follow the discussion of Mizuno and Koide [72] and start with the Hamiltonian in the form

$$\begin{aligned}
 H = & \sum_i \frac{\mathbf{p}_i^2}{2m_i} + \sum_i D_i(\mathbf{E}) + \sum_i \mathbf{f}_i(\mathbf{E}) \cdot \mathbf{r}_i + \sum_{i \leq j} \mathbf{r}_i \cdot k_{ij}(\mathbf{E}) \cdot \mathbf{r}_j \\
 & + \sum_{i < j} \mathbf{S}_i \cdot J_{ij}(\mathbf{r}, \mathbf{E}) \cdot \mathbf{S}_j + \sum_i K_i(\mathbf{S}_i, \mathbf{E}, \mathbf{r}) + \beta \sum_i \mathbf{H} \cdot g_i(\mathbf{r}) \cdot \mathbf{S}_i,
 \end{aligned} \quad (5.1)$$

where m_i is the mass, $\mathbf{p}_i$ the momentum, $\mathbf{r}_i$ the displacement from its equilibrium position, and $\mathbf{S}_i$ the spin of the i th ion. With no external electric field, the first line of the Hamiltonian (5.1) reduces to the conventional form for a lattice in the harmonic approximation and the second line to the effective spin Hamiltonian for an exchange-coupled system with exchange constants J_{ij} , single-ion anisotropy K_i , and a Zeeman term to account for MD transitions. By neglecting the latter two (single-ion) terms, assuming J_{ij} to be a scalar (isotropic, diagonal interactions), and expanding the Hamiltonian with respect to $\mathbf{r}$ and $\mathbf{E}$, one obtains at zeroth order

the well-known lattice Hamiltonian, a Heisenberg term, and a set of terms linear in $\mathbf{E}$,

$$H_E = \sum_i \mathbf{E} \cdot \left(\frac{\partial D_i}{\partial \mathbf{E}} \right)_0 + \sum_i \mathbf{E} \cdot \left(\frac{\partial \mathbf{f}_i}{\partial \mathbf{E}} \right)_0 \cdot \mathbf{r}_i + \sum_{i \leq j} \mathbf{r}_i \cdot \mathbf{E} \cdot \left(\frac{\partial k_{ij}}{\partial \mathbf{E}} \right)_0 \cdot \mathbf{r}_j \quad (5.2)$$

$$+ \sum_{i < j} \mathbf{E} \cdot \left(\frac{\partial J_{ij}}{\partial \mathbf{E}} \right)_0 \cdot (\mathbf{S}_i \cdot \mathbf{S}_j) + \sum_{i < j} (\mathbf{S}_i \cdot \mathbf{S}_j) \sum_k \mathbf{E} \cdot \left(\frac{\partial^2 J_{ij}}{\partial \mathbf{E} \partial \mathbf{r}_k} \right)_0 \cdot \mathbf{r}_k.$$

Here the second and third terms account respectively for the one- and two-phonon processes which dominate the FIR spectra in dielectric crystals. The fourth and the fifth terms contain spin-dependent effective dipole moments, and describe, respectively, dipole-active two-magnon excitation processes and two-magnon-plus-one-phonon absorptions. The former were first observed in FeF_2 [74] and MnF_2 [75] and were described theoretically by Tanabe and coworkers [10, 76]. Microscopically, the effective dipole moment of the two-magnon excitation arises from off-diagonal terms in the superexchange coupling involving two spin-flips (because of spin conservation) on two neighboring sites in the ground state [77]. In a lattice with inversion symmetry, such as the NaCl-type structure of NiO, the two-magnon excitation is not allowed. In order to break the inversion symmetry, the additional excitation of an odd-symmetry phonon is necessary, leading to two-magnon-plus-one-phonon processes which are ED-allowed [72, 78]. Observation of these processes in the parent compound of the cuprates, La_2CuO_4 , led to a generalized description of multi-magnon-plus-phonon excitations [79, 80]. As in the two-magnon process, spin-flips can also occur accompanied by the excitation of one or both of the neighboring ions into a higher orbital state, giving rise to exciton-magnon or exciton-exciton transitions, respectively [77, 81, 82]. Very recently, a hybridized magnon-phonon excitation, dubbed the electromagnon, has been discovered experimentally [83]. The electromagnon becomes observable in the presence of strong spin-lattice coupling and appears to be a feature of multiferroic systems with spiral spin structure, but the exact mechanism of the coupling is still the subject of active discussion [83, 84]. In Table 5.3, we present a short list of excitations observable by optical measurements, distinguishing between excitations related to either lattice or spin degrees of freedom and excitations arising due to coupling between the subsystems. The examples and references were chosen to include both early and recent studies and systems of differing effective magnetic dimensionality. Comprehensive review articles are indicated where available.

In the expansion of the Hamiltonian (5.1), a further term appears which does not depend on $\mathbf{E}$, but is linear in the ion displacement,

$$H_{SP} = \sum_{i < j} (\mathbf{S}_i \cdot \mathbf{S}_j) \sum_k \left(\frac{\partial J_{ij}}{\partial \mathbf{r}_k} \right)_0 \cdot \mathbf{r}_k. \quad (5.3)$$

This term describes a magnetoelastic coupling between the spins and the lattice, reflecting the influence of exchange coupling on the lattice vibrations probed by

Table 5.3 Elementary excitations detectable by optical spectroscopy in crystals with lattice, orbital, and spin degrees of freedom

Excitation	Degree of Freedom	Examples	Monographs/ Reviews
Phonon	Lattice	CdCr ₂ S ₄ [85]	[86]
One-magnon	Spins	FeF ₂ [87], LuMnO ₃ [84]	
Two-magnon	Spins	FeF ₂ [74], MnF ₂ [75], SmTiO ₃ [88]	[10]
Exciton-magnon	Spins	MnF ₂ [89], KCuF ₃ [90]	
Exciton-exciton	Spins	MnF ₂ , RbMnF ₃ [91]	
	Coupling		
Vibron	Electron–phonon	CuGeO ₃ [92], LaTiO ₃ [93]	[94]
Polaron	Electron–phonon	La _{1-x} (Ca,Sr) _x MnO ₃ [95]	[96]
Phonon-assisted	Magnon–phonon	NiO [72, 78]	[97]
Multi-magnon		La ₂ CuO ₄ [79, 80] Sr ₂ CuO ₃ [98]	
electromagnon	Magnon–phonon	TbMnO ₃ [83], TbMn ₂ O ₅ [84]	

IR spectroscopy. This term is discussed in detail for the case of the magnetically frustrated pyrochlore lattice in Chap. 11 by Tchernyshyov and Chern. In the following, we discuss experimental results for the FIR phonon spectra in Cr spinels, compounds which are the paradigm for frustrated antiferromagnets displaying this type of physics.

5.7 Spin–Phonon Interaction in Cr Spinel

The quest for realizations of highly frustrated magnets leads naturally to the spinel systems ACr_2X_4 ($X = O, S, Se$). As described in Chap. 7 by Takagi and Niitaka, in one spinel family the A site is occupied by nonmagnetic ions such as Zn or Cd [99]. The Cr^{3+} ions have spin $S = 3/2$ and are situated on corner-sharing tetrahedra which form a pyrochlore lattice (inset Fig. 5.6). Examples such as $CdCr_2O_4$ (with frustration parameter $f = \Theta_{CW}/T_N = 32$) and $ZnCr_2O_4$ ($f = 8.7$) can be regarded as paradigms for systems where the antiferromagnetic nearest-neighbor Heisenberg exchange leads to inherent frustration and considerable degeneracy of the magnetic ground state [100]. The phase diagram in Fig. 5.6 shows that the strong frustration in $ZnCr_2O_4$, where $\Theta_{CW} = -3, 98$ K, is diminished when the lattice parameter is enlarged by substituting Cd and Hg for Zn or S and Se for O. Weakening of the direct Cr–Cr coupling, which is antiferromagnetic, is accompanied by a strengthening of ferromagnetic next-nearest neighbor interactions, which lead to bond frustration with $\Theta_{CW} = T_N \approx 10$ K for $ZnCr_2S_4$, and finally to ferromagnetism in cases such as $CdCr_2S_4$. In reality, the degeneracy of the magnetic ground state is lifted at finite temperatures by the coupling to other degrees of freedom. The spin-Jahn–Teller effect is an example of this phenomenon, and

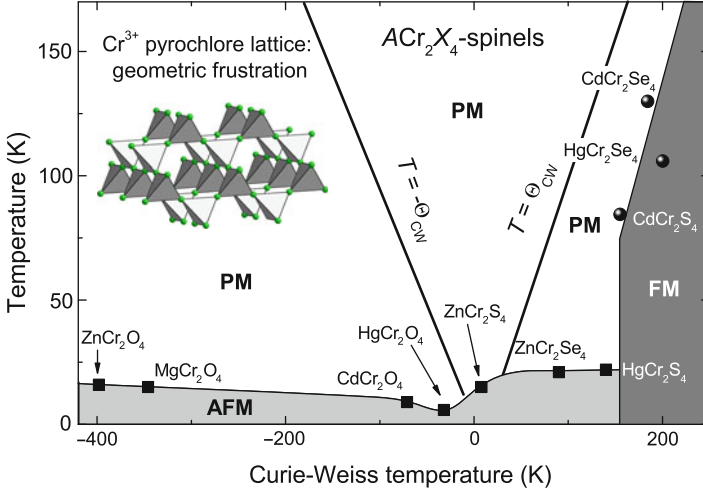


Fig. 5.6 Phase diagram for ACr_2X_4 ($A = \text{Zn, Mg, Cd}$, $X = \text{O, S, Se}$) spinels. The Cr spins occupy a pyrochlore lattice, leading to strong geometric frustration of the dominant antiferromagnetic interactions. For $X = \text{S, Se}$, competition with ferromagnetic interactions leads to a change in sign of the Curie–Weiss temperature and thus to a ferromagnetic ground state

both the single-tetrahedron case and the collective case of the pyrochlore lattice (specifically CdCr_2O_4 and ZnCr_2O_4) are presented in Chap. 11 by Tchernyshyov and Chern. This spin–Jahn–Teller effect in the Cr spinels leads to a tetragonal or an orthorhombic distortion of the lattice and to corresponding splitting of the IR-active phonon modes.

Experimentally, four phonon excitations are observed in the IR spectra of these spinel systems, which have cubic symmetry and space group $Fd\bar{3}m$ [101]. The corresponding reflectivity spectrum for CdCr_2O_4 is shown in Fig. 5.7a for the paramagnetic state at 15 K. The modes observed are consistent with the irreducible representation of the optically active phonons which can be obtained by normal mode analysis [11],

$$\Gamma = A_{1g} + E_{1g} + 3T_{2g} + 4T_{1u}. \quad (5.4)$$

In systems with inversion symmetry, the Raman and IR modes are mutually exclusive [102], i.e. modes even with respect to inversion can only be observed in a Raman experiment [103, 104] while only odd modes generate a dipole moment which allows for IR activity. Hence, the observed IR-active phonons correspond to the four triply degenerate $T_{1u}(i)$ ($i = 1, 2, 3, 4$) modes.

However, the reflectivity data at 5 K, below the magnetic ordering temperature $T_N = 8$ K, shows clearly that mode $T_{1u}(2)$ is split in the low-temperature phase (Fig. 5.7a). The temperature-dependence of its phonon frequency is shown in Fig. 5.7c and reveals a clear splitting of the mode by $\Delta\omega = 10 \text{ cm}^{-1}$ [100, 105].

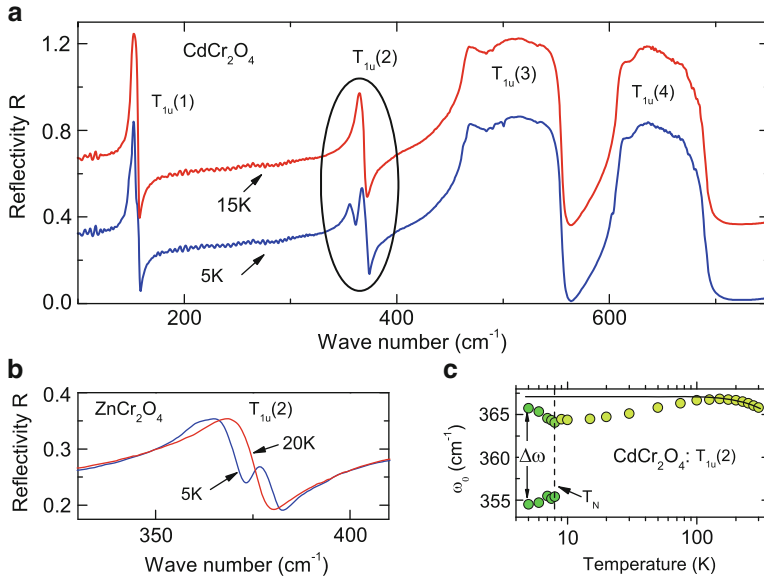


Fig. 5.7 (a) Reflectivity spectra of CdCr₂O₄ at 15 K in the paramagnetic state (shifted upwards for clarity) and at 5 K, below the antiferromagnetic ordering temperature $T_N = 8$ K. The $T_{1u}(2)$ mode is clearly split at 5 K [100]. (b) Splitting of the $T_{1u}(2)$ phonon in ZnCr₂O₄. (c) Temperature-dependence of the phonon eigenfrequency of the $T_{1u}(2)$ mode, revealing the splitting at $T_N = 8$ K in CdCr₂O₄

Similar behavior has been found for ZnCr₂O₄ (Fig. 5.7b) at the magnetostructural transition at $T_c = 12.5$ K, with a splitting $\Delta\omega = 11$ cm⁻¹ [106], and for several other Cr spinel systems [100, 107].

The magnetoelastic coupling is already manifest above the magnetostructural phase transition, because it influences the phonon frequency as soon as finite spin correlations appear. In the temperature-dependence of the phonon frequency of mode $T_{1u}(2)$ in CdCr₂O₄ (Fig. 5.7c), a deviation from the expected behavior, which would be a stiffening of the lattice vibrations due to anharmonic effects (solid line in Fig. 5.7c), is clearly visible in the softening of the phonon below 100 K, which is the order of the Curie–Weiss temperature in this compound.

By considering a nearest-neighbor Heisenberg spin system, Baltensberger and Helman related the frequency shift of lattice vibrations in magnetic crystals to the magnetic energy of the system, $\Delta\omega \approx \lambda \langle S_i \cdot S_j \rangle$: these authors introduced the spin–phonon coupling parameter λ , which depends on the derivatives of the exchange constants with respect to the coordinates of the magnetic ions [108, 109]. Information concerning the spin–phonon coupling can thus be extracted from optical measurements of phononic excitations by IR spectroscopy, if the spin–spin correlation functions can be obtained independently. By estimating the magnetic contribution from a nearest-neighbor Heisenberg term with coupling J to the specific heat C_m , the temperature-dependence of the nearest-neighbor spin–spin

correlations was extracted via

$$\langle S_i \cdot S_j \rangle = \text{const} + \frac{1}{6N_A J} \int_{T_N}^T C_m(T') dT', \quad (5.5)$$

where $6N_A$ is the number of Cr–Cr bonds per mole. Another approach is to obtain the nearest-neighbor spin–spin correlations directly from the magnetic susceptibility via

$$\langle S_i \cdot S_j \rangle = \frac{k_B T \chi(T)}{N_A g^2 \mu_B} - \frac{s(s+1)}{3}. \quad (5.6)$$

The values for ZnCr_2O_4 were found to be in the range $\lambda \approx 4.0 - 6.2 \text{ cm}^{-1}$, and $\lambda \approx 3.3 - 3.9 \text{ cm}^{-1}$ was derived in the paramagnetic regime for CdCr_2O_4 [104, 106].

The spin–phonon coupling derived from the phonon shift in the paramagnetic state should also describe the phonon splitting upon lowering the symmetry from, for example, O_h to D_{4h} at the magnetic phase transition. As a result of this symmetry-lowering, the T_{1u} triplets should split into an E_u doublet and an A_{2u} singlet [110]. The magnetostructural transition can be characterized by the spin-Peierls order parameter $n_{sp} = \langle S_1 \cdot S_2 - S_1 \cdot S_4 \rangle$ for the triply degenerate T_{1u} modes, where the spins S_i reside on the corners of each tetrahedron. The order parameter accounts for the difference in tetrahedral bonds with ferromagnetic or antiferromagnetic coupling and vanishes accordingly in the undistorted paramagnetic state. The induced phonon anisotropy for ZnCr_2O_4 in the magnetically ordered phase has been calculated from first principles [110] and the spin–phonon coupling involved in the splitting of the $T_{1u}(2)$ mode,

$$\Delta\omega \approx \lambda \langle S_1 \cdot S_2 - S_1 \cdot S_4 \rangle \quad (5.7)$$

was calculated to be $\lambda \approx 11 \text{ cm}^{-1}$. This value is about twice the one derived in the paramagnetic state with $\lambda \approx 4.0 - 6.2 \text{ cm}^{-1}$ [104, 106]. In the case of ZnCr_2O_4 , the $T_{1u}(2)$ mode reportedly splits into a lower-lying doublet and a higher singlet state [106], which is consistent with a contraction of the lattice in c -direction as illustrated in Chap. 11 by Tchernyshyov and Chern. For this distortion, which has E_g symmetry, averaging over the corresponding bonds leads to $n_{sp} = 9/2$, and the experimental splitting $\Delta\omega = 11 \text{ cm}^{-1}$ yields a spin–phonon coupling of $\lambda = 2.4 \text{ cm}^{-1}$. For CdCr_2O_4 , the doublet lies higher than the singlet [105], corresponding to an elongation along the c -axis with E_u symmetry (depicted in Chap. 11 by Tchernyshyov and Chern). The spin-Peierls order parameter is then $n_{sp} = -2.25$, whence one can estimate $\lambda \approx 4 \text{ cm}^{-1}$ from the experimentally measured splitting $\Delta\omega \approx 9 \text{ cm}^{-1}$. This value compares nicely to the one from the paramagnetic regime, justifying the assumption of a nearly collinear configuration of the spins in the ground state. Moreover, the observation of optical phonons which become both Raman and IR active [104] at the magneto-structural transition provides evidence for the loss of inversion symmetry in CdCr_2O_4 in agreement with the results of Tchernyshyov and Chern.

The reason for the discrepancy between the values derived from the splitting, $\lambda_{\text{AFM}} = 2.4 \text{ cm}^{-1}$, and from the phonon shift above T_N , $\lambda_{\text{PM}} = 4.0\text{--}6.2 \text{ cm}^{-1}$, in ZnCr_2O_4 may originate from the oversimplified assumption of such a collinear antiferromagnetic ground state: experimental studies indicate that the true low-temperature spin configuration of ZnCr_2O_4 is considerably more complex [111–113]. Hence, a value much lower than $n_{sp} = 9/2$ can be expected, which is in accordance with the fact that a complete splitting of some of the triply degenerate T_{1u} modes was reported for both ZnCr_2O_4 , confirming a lower symmetry in the magnetically ordered state [100, 104].

5.8 Exciton–Magnon Absorption in KCuF_3

The perovskite KCuF_3 is considered as a prototypical system for a cooperative Jahn–Teller effect, orbital ordering, and a quasi-one-dimensional antiferromagnetic Heisenberg chain. This is related to the particular type of orbital order in KCuF_3 , in which a single hole alternately occupies $3d_{x^2-z^2}$ and $3d_{y^2-z^2}$ orbital states of the Cu^{2+} ions ($3d^9$ electronic configuration) [114]. The cooperative Jahn–Teller distortion is characterized by CuF_6 octahedra elongated along the a and b axes and arranged in an antiferrodistortive pattern in the ab -plane. The system orders antiferromagnetically at $T_N = 39 \text{ K}$, but the spinon excitation continuum, one characteristic property of a 1D spin chain, remains observable below T_N and persists up to approximately 200 K [115].

In Fig. 5.8a, the absorption coefficient α for light polarized parallel ($\mathbf{E} \parallel c$) and perpendicular ($\mathbf{E} \perp c$) to the c -axis, the direction of the dominant antiferromagnetic coupling, is shown at 8 K . For both orientations, four broad excitation bands A_1 – A_4 are clearly visible. These absorption bands correspond to the local d - d CF excitations of the Cu^{2+} ions in a distorted octahedral environment (the level-splitting is shown as an inset in Fig. 5.8a). ED transitions between d -levels are parity-forbidden, but they can become allowed as a consequence of a perturbation of the CF potential by phonons (vibronic interaction). Here, we focus on the sharp features which can be seen at the onset of the A_2 and A_3 bands. Two sets of sidebands become visible on the enlarged scale shown in Fig. 5.8b, the respective onset energies being $8,508 \text{ cm}^{-1}$ and $9,775 \text{ cm}^{-1}$. These vibronic A_2 and A_3 transitions were ascribed to MD transitions between the CF levels, while the first three sidebands appear at separations of $88(86)$, $114(112)$, $174(175)$, and $200(203) \text{ cm}^{-1}$ from the transitions, as indicated in Fig. 5.8b. Further absorptions related to the A_2 transition can be identified at 287 , 356 , 432 , and 553 cm^{-1} .

The appearance of sidebands related to CF transitions was first reported in 1965 for MnF_2 and explained in terms of magnon sidebands arising due to an exchange-induced dipole mechanism [116, 117]. The separation in energy of the magnon sideband from the magnetic dipole transition corresponds to the magnon energy at the Brillouin zone boundary. In KCuF_3 , neutron scattering investigations reported zone-boundary energies of magnon excitations at $m_1 = 88.7$, $m_2 = 117$, and

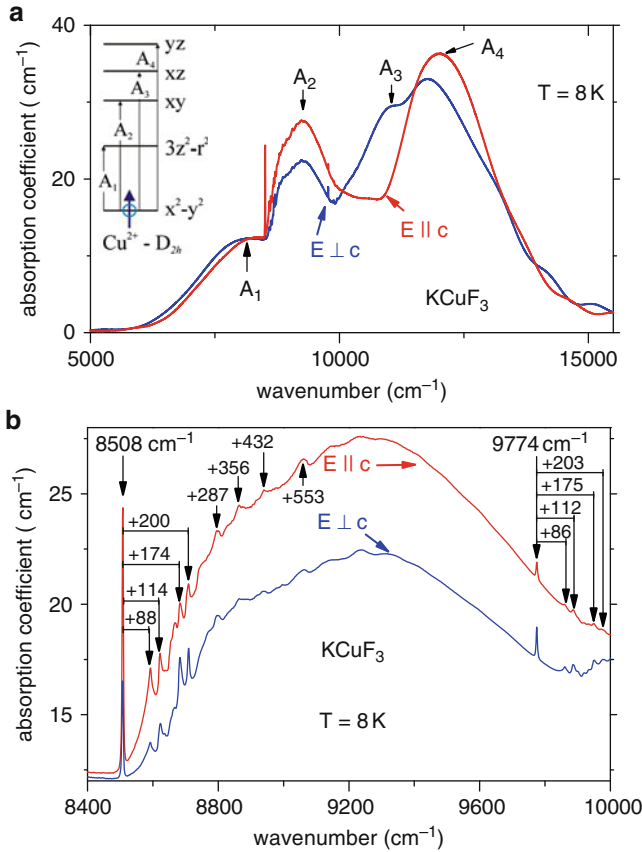


Fig. 5.8 (a) Absorption spectra of KCuF_3 at 8 K for light polarized perpendicular and parallel to the antiferromagnetic c -axis. Inset: crystal-field (CF) level-splitting of Cu^{2+} in D_{2h} symmetry. (b) Fine structures related to the CF transitions A_2 and A_3 for $E \parallel c$ and $E \perp c$ at 8 K. Frequency shifts with respect to the MD lines at 8,508 and 9,774 cm^{-1} are indicated

$m_3 = 221 \text{ cm}^{-1}$ [118, 119]. The first and second sideband peaks are in good agreement with the first two magnon energies, the third sideband can be assigned to a two-magnon process involving the 88.7 cm^{-1} magnon, and the fourth corresponds to a two-magnon process involving the 88.7 and the 117 cm^{-1} magnons. The further sidebands related to A_2 also agree excellently with $2m_1 + m_2 = 294.4 \text{ cm}^{-1}$, $4m_1 = 354.8 \text{ cm}^{-1}$, $m_1 + 3m_2 = 439.7 \text{ cm}^{-1}$, and $m_1 + 4m_2 = 556.7 \text{ cm}^{-1}$. The fact that five or more magnon sidebands can be observed with such clarity is not yet understood, but it may be connected with the coexistence of spinon and magnon excitations in this compound [115]. The temperature-dependence of the MD transition at $8,508 \text{ cm}^{-1}$ and of the magnon sidebands is displayed in Fig. 5.9 for $E \parallel c$. The fine structure emerges below a temperature $T_S = 50 \text{ K}$, while the integrated intensity of the MD transition, shown as an inset in Fig. 5.9, displays

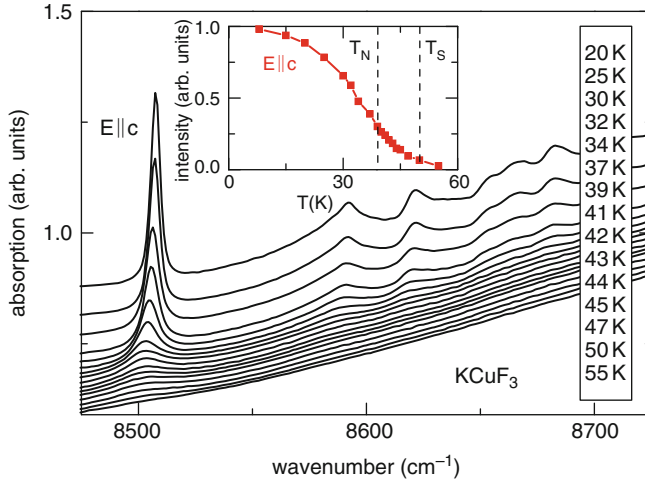


Fig. 5.9 Temperature-dependence of the exciton-magnon bands below $T_S = 50$ K for $E \parallel c$. Inset: temperature-dependence of the intensity of the zero-magnon line for $E \parallel c$. The solid line is a guide to the eye

no visible anomaly at $T_N = 39$ K. This temperature has been suggested as the onset of short-range order arising as the result of a freezing-in of dynamical lattice distortions [120].

This example shows that optical spectroscopy is a very sensitive probe of short-range-order effects and finite spin correlations above the magnetic ordering temperature. Hence, optical measurements of spin-forbidden local orbital transitions can provide important information on the spin-excitation spectrum both at low temperatures and in the correlated paramagnetic phase of frustrated magnets.

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